SURYA K

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Education

Vellore Institute of Technology, Chennai *Jul 2024 – May 2026*
Master of Computer Applications CGPA: 9.10/10.0

PSG College of Technology, Coimbatore *Sep 2021 – May 2024*
Bachelor of Computer Systems and Design CGPA: 8.23/10.0

Venkatalakshmi Matriculation Higher Secondary School, Coimbatore *May 2021*
HSC, State Board of Tamil Nadu Percentage: 93.16%

Technical Skills

- **Programming Languages:** Python, Java, C++, SQL
- **Data Analytics:** Power BI, Excel, Data Analysis, EDA, Data Visualization (Matplotlib, Seaborn, Plotly)
- **ML & AI:** Machine Learning, Deep Learning, Generative AI, Agentic AI, Federated Learning, Transfer Learning, Predictive Modeling, Feature Engineering, Model Evaluation, Hypothesis Testing, Supervised & Unsupervised Learning (Classification, Regression, Clustering and Association)
- **Web Technologies:** Django REST Framework, React, Node.js, REST APIs
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, PennyLane
- **Tools:** Git, GitHub, Jupyter Notebook

Projects

Noise-Aware Hybrid Quantum Transfer Learning for Breast Cancer Diagnosis *2025*
Python, PyTorch, PennyLane, OpenCV, NumPy

- Developed a hybrid quantum-classical deep learning model for breast cancer classification using the BreakHis histopathology dataset.
- Integrated EfficientNet-B0 with a variational quantum circuit to enhance feature extraction and classification performance.
- Designed data preprocessing and augmentation pipelines for histopathology images to improve model generalization.
- Achieved **90% classification accuracy** and evaluated robustness under Pauli, phase-damping, and depolarizing noise using Zero-Noise Extrapolation (ZNE).
- Assessed performance using ROC-AUC, F1-score, sensitivity, specificity, and generated visualizations to compare model behavior across noise settings.

FedAdapt: Federated Learning with Riemannian Manifold & ATCNet for EEG Classification *2025–2026*
Python, PyTorch, BCI, Flower, MOABB, ATCNet

- Developed a federated learning framework for motor imagery EEG classification with decentralized client training.
- Applied Riemannian covariance feature extraction and the ATCNet architecture to improve learning from non-IID EEG data.
- Built preprocessing and training pipelines using the BCI Competition IV 2a dataset with MOABB.
- Implemented client-server communication, model aggregation, and evaluation using the Flower federated learning framework.
- Achieved **78.8% cross-subject accuracy** while preserving data privacy through collaborative model training.

AI-Powered Expense Tracker & Financial Planning System *2026*
Django REST Framework, React, SQLite, REST API, NLP

- Built a full-stack expense management application using Django REST Framework, React, and SQLite with secure token-based authentication.
- Developed REST APIs for expense management, user authentication, analytics, and AI-powered financial insights.
- Implemented NLP-based parsing to convert natural language expense entries into structured financial records.
- Designed interactive dashboards for expense trends, category-wise analysis, budgeting, savings tracking, and financial health monitoring.
- Integrated frontend and backend components to provide real-time data visualization and a seamless user experience.

Certifications

- **Data Foundations Associate**, Oracle
- **AI Engineer for Developers Associate**, Datacamp

Leadership & Initiatives

- **Placement Representative**, B.Sc. CSD Batch, PSG College of Technology (2021–2024): Coordinated between 60+ students and the placement cell, facilitating campus recruitment drives and career readiness sessions.
- **Convenor & Core Member**, Animal Welfare Club, PSG College of Technology (2022–2024): Led a team of 20+ volunteers to organize campus awareness campaigns and community outreach events.
- **ERM Lead**, PSG College of Technology (2023–2024): Managed cross-functional teams for departmental fests, overseeing logistics, scheduling, and on-ground execution of 8 events that year.